

Pointing jitter correction in hollow-core fiber nonlinear systems based on LSTM

KaiXuan Li,^{1,2,†} HongYang Li,^{1,2,3,†} LiWei Song,^{1,2,*}  AND YE TIAN^{1,2,4} 

¹State Key Laboratory of Ultra-intense Laser Science and Technology, Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghai 201800, China

²Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing 100049, China

³School of Physics Science and Engineering, Tongji University, Shanghai 200092, China

⁴tianye@siom.ac.cn

[†]These authors contributed equally to this work.

*slw@siom.ac.cn

Received 16 January 2025; revised 21 April 2025; accepted 28 April 2025; posted 28 April 2025; published 15 May 2025

Hollow-core fiber has been widely used for laser spectrum broadening and pulse duration compression, and its performance is highly sensitive to pointing instability caused by environmental interference and system nonlinearity. Here, we propose a feedback control system based on a three-layer long short-term memory network for real-time pointing jitter correction, which reduces the beam pointing jitter from 4.04 and 4.64 μrad to 0.869 and 0.903 μrad in the horizontal and vertical directions, respectively. Our work demonstrates the application of machine learning in pointing stabilization of nonlinear systems and paves the way for highly stable few-cycle laser pulse generation. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

<https://doi.org/10.1364/AO.557151>

1. INTRODUCTION

Femtosecond lasers play a crucial role in modern science, military, and industrial applications [1,2], with femtosecond laser pointing stability being a key parameter for ensuring optimal system performance [3–5]. Pointing jitter in femtosecond lasers is primarily caused by factors such as thermal drift, mechanical vibrations, measurement errors, air turbulence, and human operations [6,7]. These factors severely affect measurement accuracy, signal quality, and system reliability. This is especially critical in high-precision applications, including air filament transmission [8,9], laser acceleration, radiation sources [10–12], and micro-nano machining [13]. Effectively correcting femtosecond laser pointing jitter is essential for ensuring the accuracy and stability of these systems. In linear systems, femtosecond laser pointing jitter can be effectively corrected using a Gaussian mixture model combined with Gaussian mixture regression, reducing the jitter to approximately 0.23 μrad [6]. For nonlinear optical systems, the interaction of nonlinear effects, temperature gradients, atmospheric turbulence, and mechanical vibrations makes beam position prediction and correction considerably more difficult. Furthermore, the input–output relationship in nonlinear optical systems is inherently complex, posing significant challenges for traditional correction methods such as mechanical stabilization or optical feedback systems, which often fail to effectively address the intricate and dynamic nonlinearities involved [14]. Therefore,

there is an urgent need to develop new methods for stabilizing femtosecond laser pointing in nonlinear systems, particularly in environments characterized by complex and dynamic disturbances.

Gas-filled hollow-core fiber (HCF) has been intensively applied for laser spectrum broadening via self-phase modulation and cross-phase modulation [15–19]. Strong nonlinear processes in HCF result in the output spectrum and transmission efficiency being highly dependent on the pointing stability, which is susceptible to mechanical vibrations, airflow, and thermal drift. In addressing the issue of laser beam pointing stability, beam stabilizers consisting of beam position detectors, actuated mirrors, and a linear feedback loop have been widely equipped in HCF systems [20]. Compensating for complex nonlinear disturbances with linear feedback control systems is challenging. However, machine learning algorithms [21], particularly long short-term memory (LSTM) networks, known for their ability to handle long-term dependencies [22], offer a promising approach to predicting and correcting laser pointing jitter induced by environmental vibrations and associated nonlinear effects. By learning and adapting to complex environments in real time, LSTM networks can improve the beam pointing stability of HCF systems, compensating for pointing deviations caused by nonlinear processes and enhancing their applicability in experiments requiring high precision. Notably, the adaptive capability of LSTM networks enables real-time optimization in

changing environments, improving the system's response speed and accuracy [23].

In this work, we present an LSTM-based control system for real-time correction of femtosecond laser pointing jitter in HCF nonlinear systems. The developed system uses a three-layer LSTM network to predict and compensate for pointing deviations, significantly reducing the root mean square (RMS) of pointing errors. Through comparison and analysis, the performance of the LSTM-based control system outperforms traditional methods and some other networks. The RMS of pointing jitter in both horizontal and vertical directions decreased from 4.04 and 4.64 μrad under free-running conditions to 0.8690 and 0.9038 μrad , respectively. This advancement demonstrates the potential of LSTM-based control systems in compensating for the nonlinear dynamics of HCF systems, providing the possibility of integrating them into advanced optical devices where precision and stability are critical.

2. EXPERIMENTAL SETUP AND RESULTS ANALYSIS

As shown in Fig. 1, a Ti:sapphire amplifier delivers laser pulses with a 50 fs pulse duration, 1.6 mJ pulse energy, and an 800 nm central wavelength at a repetition rate of 1 kHz. To address the high sensitivity of the HCF to incident light pointing jitter, a beam stabilization system based on fast steering mirrors (FSMs) is implemented before and after the fiber. The system consists of two pairs of FSMs, a position sensitive detector (PSD), and an analysis and control unit. The laser pulses are first stabilized by the pointing stabilization system consisting of FSM1 and FSM2. Approximately 5% of the laser beam is then directed to the PSD via BS1 to monitor real-time beam position changes before the fiber. The PSD transmits the beam position data to the analysis and control unit, which quickly processes the signal and generates feedback control signals to drive the FSMs, adjusting the femtosecond laser's pointing in real time. FSM3 and FSM4 are used for the beam alignment before chirped mirrors, which are not involved in the current feedback system. In the future, we will upgrade the machine learning algorithms to achieve synchronized control of the beam stabilization systems, aiming to further improve the output beam stability and provide optimal laser conditions for subsequent experiments. A dual-channel PSD is employed to measure the beam pointing reflected by BS1 and BS2 at the same time. This configuration enables simultaneous dual-point detection within a single device, ensuring data correlation and high temporal resolution. These two data channels are used as the inputs of the LSTM model, respectively, enhancing the accuracy and responsiveness of the feedback control system.

As a result, the RMS of pointing jitter in both the horizontal and vertical directions of the femtosecond laser focal plane is 2.8 and 3.0 μrad , respectively. Then, the mean beam with 1.5 mJ pulse energy is coupled into a gas-filled HCF with 1 m length and 400 μm core diameter by a lens with a 1.5 m focal length. The laser spectrum is broadened via the self-phase modulation (SPM) effect under appropriate neon gas pressure. After dispersion compensation by chirped mirrors, few-cycle laser pulses with a duration of 10 fs and an energy of approximately 1 mJ are achieved. To evaluate the pointing stability under free-running

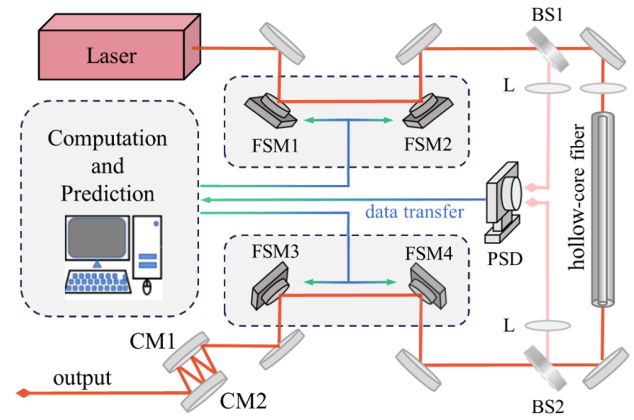


Fig. 1. Schematic illustration of the experimental setup. FSM1–4, fast steering mirrors; BS1,2, beam splitters with a reflectivity of 5%; L, lens; hollow-core fiber, HCF with 1 m length and 400 μm core diameter; CM1, 2, chirped mirrors; PSD, position sensitive detector.

conditions after the HCF, 5% of the output beam is reflected by BS2. The dynamic changes in the centroid distribution of the far-field spot on the focal plane are then recorded by a PSD.

As shown in Figs. 2(a) and 2(b), the free-running RMS errors of pointing jitter in the horizontal (X) and vertical (Y) directions are 4.04 and 4.64 μrad , respectively. The heatmaps in these figures represent the distribution of pointing jitter along the horizontal and vertical directions. Figure 2(c) shows the pointing jitter distribution during the measurement. As shown in Fig. 2(d), the R actual represents the distance between the point and the center position, and is calculated according to the formula: $R = \sqrt{X^2 + Y^2}$, where X and Y are the horizontal and vertical positions, respectively.

After the HCF, the laser beam pointing jitter becomes more pronounced, which is mainly caused by nonlinear effects within HCF, as well as unavoidable noise, such as perturbations due to mechanical vibrations and air turbulence. As the pointing variations are highly nonlinear and strongly correlated with time-series information, it is difficult to achieve ideal control with traditional linear feedback methods. LSTM is well-suited for solving sequential and time-series problems due to its ability to retain memory context within the units. Additionally, it can effectively overcome challenges such as exploding gradients or vanishing gradients during backpropagation. Compared with other algorithms, LSTM networks demonstrate superior performance in long-term predictions; their flexibility provides an optimal solution for improving the performance of fiber systems.

In the experiment, a three-layer LSTM network is designed to predict the pointing position in real time after the fiber. Each LSTM node contains three gates: the forget gate (f_z), the input gate (i_z), and the output gate (o_z), which are represented as follows:

$$\begin{aligned} f_z &= \sigma(W_f[H_{z-1}, X_t] + b_f) & \tilde{C}_z &= \tanh(W_c[H_{z-1}, X_t] + b_c) \\ i_z &= \sigma(W_i[H_{z-1}, X_t] + b_i) & o_z &= \sigma(W_o[H_{z-1}, X_t] + b_o) \\ C_z &= f_z \cdot C_{z-1} + i_z \cdot \tilde{C}_z & H_z &= o_z \cdot \tanh(C_z), \end{aligned} \quad (1)$$

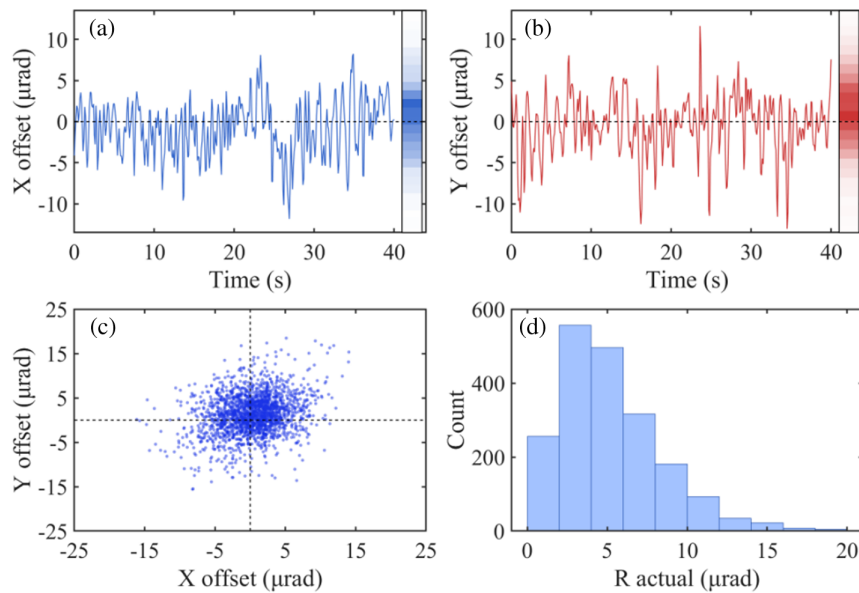


Fig. 2. Horizontal (X axis) (a) and vertical (Y axis) (b) pointing stability without correction. Schematic diagram of pointing jitter distribution (c) and center point statistics (d).

where σ and $\tanh$ are the activation function and the hyperbolic tangent function, respectively; X_t represents the input value; z denotes the current time step; $z-1$ refers to the previous time step; H_{z-1} is the previous cell output state; C_z represents the internal candidate cell state; W_f , W_i , W_o , W_c denote the weight matrices; b_f , b_i , b_o , b_c are the bias parameters. Our LSTM network is designed based on Eq. (1) to effectively capture temporal dependencies in laser beam pointing data. The forget gate discards irrelevant past information, which prevents noise accumulation. The input gate selectively updates the memory cell with new beam jitter data. The cell state stores historical trends in beam pointing drift, allowing the model to make long-term predictions. The output gate determines how much information contributes to the current prediction.

The design of the LSTM network involves four steps: data preprocessing, network design, training, and validation. During data preprocessing, input sequential data are normalized to ensure all features are on the same scale, allowing the network to learn effectively. By increasing the depth of the hidden layers, the three-layer LSTM enables deeper and more sophisticated learning, resulting in significantly improved prediction accuracy and optimization performance. After a series of experiments and parameter adjustments, the final LSTM network architecture is determined, as shown in Fig. 3. We employed a three-layer LSTM network with 64 nodes in each layer, and the network was trained with a batch size of 64 over 70 epochs. This mode demonstrated optimal predictive performance and training stability in multiple experiments. Choosing a three-layer network strikes a balance between model complexity and training efficiency. Single-layer and two-layer LSTM networks were prone to local minima, resulting in suboptimal predictions. In contrast, the three-layer structure effectively captures complex temporal dependencies, enhancing predictive accuracy. Adding more layers (such as four or five) did not significantly improve performance and increased training time and computational

resource consumption. Therefore, the three-layer structure was deemed optimal.

Regarding the number of nodes, we tried configurations with the same number of nodes per layer and with a reduced number of nodes per layer. The configuration with 64 nodes per layer yielded the best predictive performance and training stability. Selecting 64 (a power of 2) nodes per layer can improve computational efficiency, especially in parallel computing environments. Additionally, the loss function stabilized after approximately 20 epochs during training, indicating model convergence, as shown in the lower right figure of Fig. 3. For the additional parameters, we chose a learning rate of 0.01, MSE as the loss function, and the Adam optimizer, all of which provide a good balance between convergence speed, stability, and performance. Despite encountering local optima in its loss curve, the three-layer LSTM network leveraged its deeper structure and enhanced gradient propagation capability to effectively overcome the challenge. This allowed the network to explore better solution spaces.

The pointing jitter correction process involves three key steps. First, preliminary beam data are collected, followed by system setup and parameter adjustments to ensure accurate measurements. A three-layer LSTM network is designed and pre-trained. Next, the data from the actual HCF system are used to further train the network. Finally, the system utilizes real-time beam data from the HCF to monitor, predict, and optimize laser pointing. Through continuous training and adjustments, the system rapidly drives the fast-steering mirrors to effectively correct pointing jitter.

Figures 4(a) and 4(b) illustrate the horizontal (X) and vertical (Y) pointing jitter predictions of the femtosecond laser over a 20 s interval using the system. From the figures, a high degree of consistency can be observed between the actual values (dark blue dots) and the predicted values (orange lines), demonstrating the network's effectiveness in capturing the trends in beam pointing

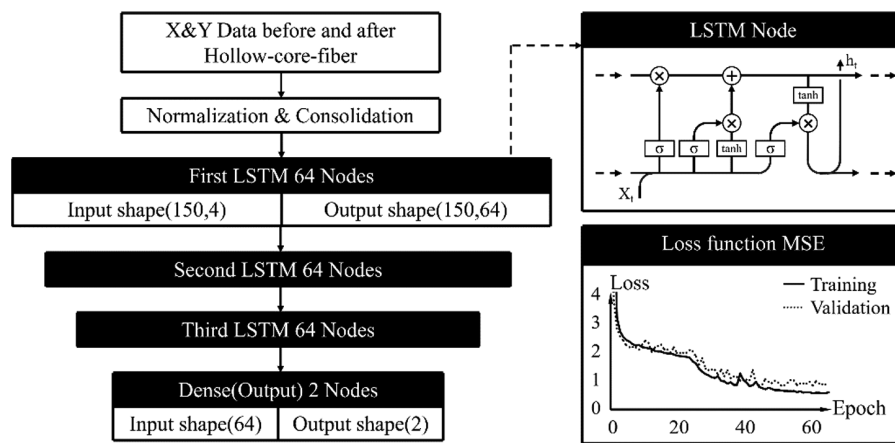


Fig. 3. LSTM network structure diagram.

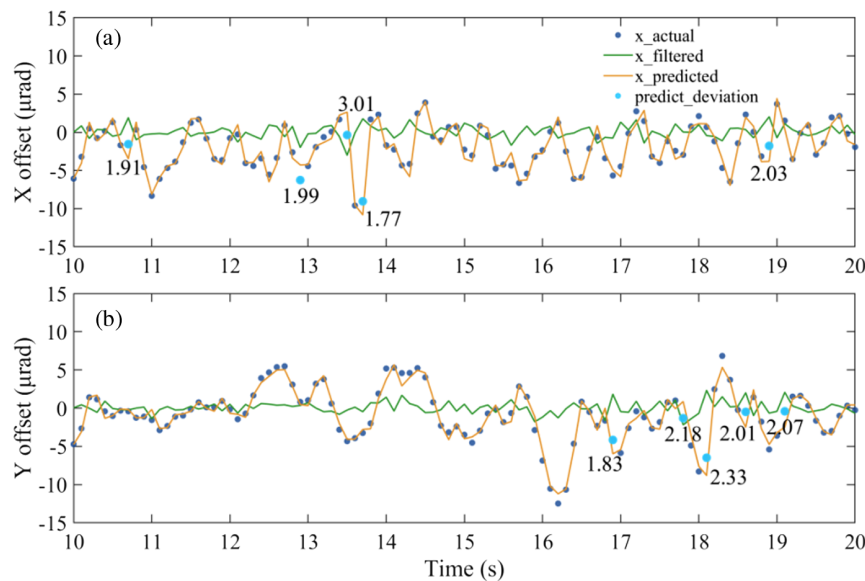


Fig. 4. Schematic illustration of predicted horizontal (a) and vertical (b) pointing stability.

variations. The x_{filtered} curve (green line) represents the deviation between the measured beam pointing and the predicted value, effectively reflecting the optimized beam trajectory after applying the correction. The predicted values closely follow the fluctuations of the actual values, further validating the network's adaptability in handling time-series data. This provides reliable support for laser pointing jitter correction. The predicted deviation (blue dots) marks a few selected points where the predicted values differ significantly from the actual values. These points are part of the $x_{\text{predicted}}$ series and are labeled specifically to illustrate the potential maximum prediction error. These points typically occur during periods of rapid and irregular fluctuations in the actual data, where the difference between consecutive points is large and lacks a predictable pattern. Since the LSTM network heavily relies on prior time-step information and is affected by the nonlinear of the HCF, it may generate relatively large prediction errors under such conditions. The results highlight the strong capabilities of the LSTM network in real-time monitoring and rapid prediction.

Under free-running conditions, the femtosecond laser beam exhibits significant pointing instability, with substantial fluctuations observed in both the horizontal and vertical directions. The scatter plot reveals a random distribution of offset points, and the R-value distribution is widely spread. After optimization, the RMS values were reduced to 0.8690 and 0.9038 μrad , respectively, as shown in Figs. 5(a) and 5(b). The optimized pointing jitter distribution became concentrated in the central region, as illustrated in Fig. 5(c). The R-value distribution also converged significantly within a smaller range, as shown in Fig. 5(d), indicating a substantial reduction in femtosecond laser pointing errors and a marked improvement in beam precision and stability. The smooth data segments near the 40 s are frequency heatmaps, where the depth of color expresses the distribution frequency of the data. They are independent visualizations and are not part of any time-series curve. Frequency heatmaps show the distribution pattern of data segments, which are mainly concentrated near zero. The LSTM network not only captured the system's dynamic behavior and complex nonlinear characteristics but also effectively enhanced system stability,

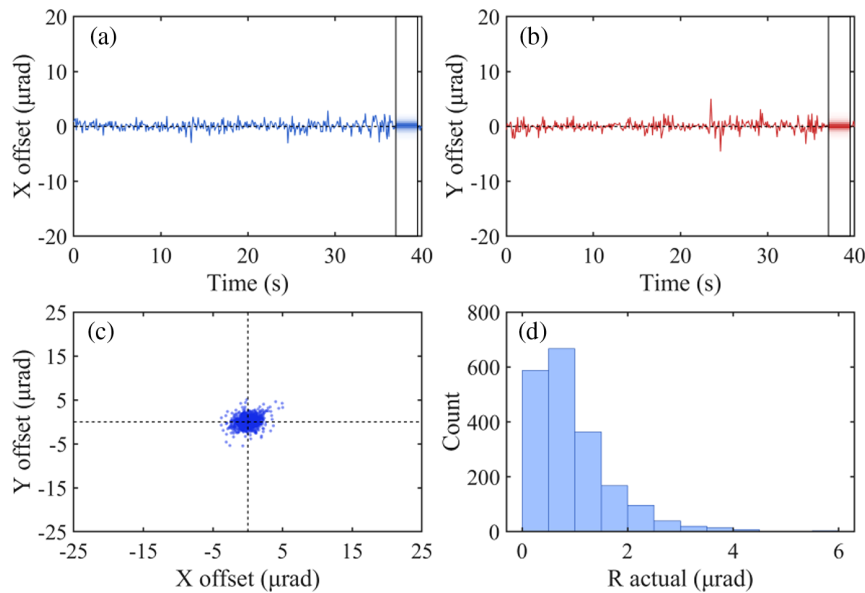


Fig. 5. Horizontal (X axis) (a) and vertical (Y axis) (b) pointing stability with correction. Schematic diagram of pointing jitter distribution (c) and center point statistics (d).

Table 1. Optimization Effects of Different Systems

Method	Information	MSE	RMS(x)	RMS(y)	$\bar{R}$
Raw data	—	16.3208 (var)	4.0399	4.6388	5.2451
LSTM	3 layers 64 nodes 70 epochs	0.7860	0.8690	0.9038	0.9785
LSTM	2 layers 64 nodes 70 epochs	0.9897	0.9844	1.0051	1.0426
LSTM	1 layer 64 nodes 70 epochs	2.2795	1.6175	1.3938	1.7030
Linear	Flatten 70 epochs	2.8589	1.7251	1.6558	2.0207
Lasso	$\lambda = 0.01$ 70 epochs	2.8648	1.7359	1.6481	2.0137
Random forest	n_estimators = 100	5.806	2.4305	2.3987	2.8448
FCNN	3 layers 16 nodes 70 epochs	7.5006	2.6528	2.8206	3.3588

demonstrating its powerful predictive and corrective capabilities in addressing pointing jitter caused by nonlinearities in the HCF system.

Table 1 presents the optimization performance of various methods in femtosecond laser pointing control, including different network architectures and their corresponding performance. While the autoregressive integrated moving average (ARIMA) model performs well in short-term predictions, it struggles with complex nonlinear relationships and long-term dependencies. Support vector regression (SVR) shows moderate performance on nonlinear problems but has limited adaptability to high-dimensional data. The results indicate that the three-layer LSTM network performs the best in dynamic environments. In contrast, while single-layer and two-layer LSTM networks improve prediction accuracy to a certain extent, they fall short of the three-layer LSTM in modeling complex features and capturing long-term dependencies. Traditional methods, such as linear regression, Lasso regression, and random forest, show clear limitations when addressing nonlinearities and time-series dependencies, failing to match the optimization performance of LSTM networks. FCNN, namely fully connected neural network, is a basic multi-layer model in which each node only performs linear operations and has no memory function. It serves as a baseline to highlight the memory advantage of

LSTM networks in time-series forecasting. Furthermore, the experimental results emphasize that the three-layer LSTM not only achieves higher accuracy in capturing nonlinear system characteristics but also exhibits excellent convergence properties. Despite encountering local optima during training, the deep network's strong learning capabilities effectively overcame the challenge.

In feedback control systems, the feedback bandwidth is a key metric for evaluating system responsiveness. In our experiments, the optimized LSTM model includes data acquisition, preprocessing, model prediction, and control signal output. Since the model calculation speed is faster than the optical and mechanical response, the maximum correction frequency of the system is currently limited by the actuator bandwidth of the laser beam stabilization system, which is 100 Hz. With the introduction of higher-speed data acquisition cards and actuators in the future, the LSTM model can achieve a correction frequency of several kHz or even tens of kHz.

3. CONCLUSION

In conclusion, we proposed an LSTM-based feedback system to predict and correct the pointing jitter of HCF nonlinear systems. A three-layer LSTM network is developed and trained

using experimental data collected from the HCF system. Experimental results showed that the RMS pointing errors in the horizontal and vertical directions were significantly reduced from 4.04 and 4.64 μrad in the free-running state to 0.8690 and 0.9038 μrad , respectively. Compared with traditional approaches and other networks, the proposed network demonstrated superior performance. This reduction highlights the network's ability to accurately predict and compensate for pointing jitter. The LSTM-based system not only improved stability and accuracy but also proved highly adaptable to dynamic environments while maintaining computational efficiency. This method demonstrates the potential of machine learning (particularly LSTM) in the real-time optimization of optical systems, providing a robust solution for maintaining precision in dynamic and complex scenarios. The work not only paves the way for precise stabilization of broadband spectrum and few-cycle laser pulses generated by HCF but also establishes a solid foundation for applying advanced predictive models in other high-precision optical and fiber systems.

Funding. National Natural Science Foundation of China (12325409, 12388102, U2267204, U23A6002); National Key Research and Development Program of China (2022YFA1604401); CAS Project for Young Scientists in Basic Research (YSBR-059, YSBR-060); Shanghai Pilot Program for Basic Research—Chinese Academy of Sciences, Shanghai Branch.

Acknowledgment. The authors acknowledge Dr. Meng Pang and Dr. Zhiyuan Huang for providing the HCF testing platform, and Mrs. Yinuo Zhao and Mr. Ning Zhao for indispensable technical support.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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